

CS3 Case Study Rubric: Sentiment Analysis of Restaurant Reviews

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Formatting	• Clear organization with labeled sections • Concise and readable writing • Code runs without errors • All required components are included • Includes clear explanation of results in written form
Task Overview	Goal: • Build a model that classifies restaurant reviews as positive, neutral, or negative using only the text of the reviews • Evaluate model performance, compare the two models, and explain key findings in a short written summary
Data Preparation	Goal: Understand and verify the dataset used for modeling • Use the provided cleaned dataset • Understand how sentiment labels are assigned in the dataset • Check for missing or invalid values in the dataset Meets Expectations: • Dataset is correctly loaded • Student verifies that the data is usable • Student demonstrates understanding of how the data is structured
Feature Creation	Goal: Convert text into numerical features • Use TF-IDF to represent review text • Apply consistent feature settings across the dataset ◦ (e.g., same vocabulary and parameters for all data) Meets Expectations: • TF-IDF is implemented correctly • Features are appropriate for model input
Modeling	Goal: Train a model to classify sentiment • Use classification models (Logistic Regression and Naive Bayes) • Split data into training and testing sets • Understand how class imbalance is addressed (e.g., through class weighting) Meets Expectations: • Model is trained correctly • Training/testing split is applied • Model choice is appropriate
Evaluation	Goal: Assess model performance • Report and compare accuracy on test data for both models • Examine performance across sentiment classes using confusion matrices and classification reports

	● Identify strengths and weaknesses of the model Meets Expectations: ● Accuracy is calculated correctly ● Performance is interpreted (not just reported) ● Limitations are acknowledged ● Student correctly interprets confusion matrices and classification reports
Interpretation & Insights	Goal: Explain what the model reveals ● Discuss patterns in predictions ● Compare the performance of the two models ● Identify challenges (e.g., difficulty classifying neutral sentiment) ● Explain impact of class imbalance ● Reflect on why the model performs as it does Meets Expectations: ● Insights go beyond surface-level ● Clear explanation of results ● Demonstrates understanding of model behavior
Success Criteria	● A strong model achieves ~70% accuracy or higher ● The full pipeline (data understanding → features → model → evaluation) is completed ● The student provides a clear explanation of results, model comparison, and limitations ● The student demonstrates understanding of both model performance and its limitations Deliverables ● Students must submit a completed Jupyter Notebook and a short written summary (approximately 1–2 pages) as a separate PDF document ● The summary should include: ○ Model performance comparison ○ Key insights and interpretation ○ Limitations of the model